TENSORING WITH A K-FLAT COMPLEX PRESERVES QUASI-ISOMORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: it names both the reformulation of K-flatness and the cone characterisation of quasi-isomorphisms.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules. Same chapter, the definition of K-flatness and its reformulation as exactness of the derived tensor product functor with the complex. Supplies the reformulation in which the statement is proved.
- Total tensor product of complexes of O-modules. Same chapter, the total complex Tot of a double complex and its functoriality in each variable. Supplies the functor whose behaviour on quasi-isomorphisms is asserted.
- Cone of a morphism and the characterisation of quasi-isomorphisms by acyclic cones. Same chapter, the mapping cone construction and the fact that a morphism of complexes is a quasi-isomorphism exactly when its cone is acyclic. Supplies the passage from preservation of quasi-isomorphisms to preservation of acyclicity, which is what the cited lemma delivers.

Source reference. Original lemma 06YM: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2969–2969. Same-chapter navigation pointer invoked at proof lines 3027-3028, where the author reduces the statement to exactness of the derived tensor product functor for a K-flat complex. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{K}^\bullet$ be a K-flat complex. Then the functor*

$$K(Mod(\mathcal{O})) \longrightarrow K(Mod(\mathcal{O})), \quad \mathcal{F}^\bullet \longmapsto Tot(\mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{K}^\bullet)$$

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transforms quasi-isomorphisms into quasi-isomorphisms.

Proof. Follows from Lemma 06YM and the fact that quasi-isomorphisms are characterized by having acyclic cones. □